

Exact directional signalling for Ising, partial-SWAP, and XY unitary families

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Draft of 30 July 2026

Abstract

For a bipartite unitary U on $A \otimes B$, directional signalling from B to A is the diamond-norm distance

$$\delta_A(U) = \inf_{\mathcal{E} \in \text{CPTP}} \|\text{Tr}_B \circ \text{Ad}_U - \mathcal{E} \circ \text{Tr}_B\|_{\diamond}.$$

This quantity was introduced by Barsse, Perinotti, Tosini, and Vaglini. Exact values were subsequently obtained for the CNOT and SWAP gates, but to our knowledge continuous interacting families have not been evaluated. For the two-qubit Ising family $U_{\theta} = \exp(-i\theta Z \otimes Z)$, we first prove $\delta_A(U_{\theta}) = |\sin 2\theta|$ and identify an optimal dephasing channel. This elementary baseline fixes conventions and extends the known CNOT endpoint. Our first main result concerns the partial-SWAP family on $\mathbb{C}^d \otimes \mathbb{C}^d$, $U_{\phi} = \cos \phi I - i \sin \phi \text{SWAP}$, where unitary covariance reduces the optimization to one depolarizing parameter λ . We evaluate the fixed- λ diamond norm in closed form and thereby reduce the full problem to a scalar convex minimization. In the weak-interaction regime $0 \leq \sin \phi \leq (d^2 - 3)/(d^2 - 1)$, the result simplifies to $\delta_A(U_{\phi}) = 2 \sin \phi$; beyond this dimension-dependent threshold the optimal effective channel moves away from the identity, reaching the known SWAP value $2(1 - 1/d^2)$. Our second main result treats the less symmetric XY line $e^{-i\theta(XX + YY)}$. Axial covariance gives an exact two-parameter min-max formula. Writing $S = \sin 2\theta$, the weak branch is $\sin 4\theta$, while near iSWAP the value is $S + S^2/2$, reaching the global two-qubit ceiling $3/2$ at iSWAP. In the intervening branch the value is the unique physical root of an explicit quartic. The proofs include matching operational witnesses: explicit input states on the closed branches, and a convex-concave saddle on the intermediate one. We also distinguish these diamond-norm results from Hilbert-Schmidt operator-entanglement measures and provide exact-arithmetic computational certificates.

Keywords: directional signalling; diamond norm; partial SWAP; iSWAP; Ising interaction; covariant quantum channels.

1 Introduction

An interacting microscopic evolution need not induce autonomous dynamics on a subsystem. Given a unitary channel Ad_U on $A \otimes B$, one may ask how well the reduced output on A can be simulated from the reduced input on A alone. The operational answer is

$$\delta_A(U) = \inf_{\mathcal{E} \in \text{CPTP}(A)} \|\text{Tr}_B \circ \text{Ad}_U - \mathcal{E} \circ \text{Tr}_B\|_{\diamond}. \quad (1)$$

The optimization includes arbitrary AB correlations and arbitrary reference systems through the diamond norm.

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Equation (1) is not a new measure. After exchanging subsystem labels, it is exactly the directional-signalling quantity introduced in Ref. [1]. In their notation, signalling from A to B' is

$$S_{A \rightarrow B'}(\mathcal{U}) = \inf_{\mathcal{C}} \|\text{Tr}_{A'} \circ \mathcal{U} - \mathcal{C} \circ \text{Tr}_A\|_{\diamond}.$$

Thus Eq. (1) is $S_{B \rightarrow A'}(\text{Ad}_U)$: exchange their A, A' with our hidden B, B' and their B, B' with our visible A, A' . Both definitions use the standard, unhalved diamond norm, so there is no normalization factor. Ref. [1] gives $S(\text{CNOT}) \leq 1$ as part of proving a strict separation from causal influence. The follow-up Ref. [2] proves the matching lower bound and hence the exact equality $S(\text{CNOT}) = 1$; it also treats SWAP, including parallel and asymptotic uses, and gives the qudit SWAP value used below. It further proves the universal bound

$$\delta_A(U) \leq 2 \left(1 - \frac{1}{d^2}\right) \quad (2)$$

for equal local dimension d , with equality at SWAP, and uses covariance to identify the completely depolarizing effective channel there. Our partial-SWAP endpoint uses that precedent; the contribution is the continuous family and its threshold. Earlier work characterized causal and semicausal operations structurally [3, 4]. In the present notation, $\delta_A(U) = 0$ is exactly semicausality from B to A : the reduced output on A then factors through the reduced input on A . Continuity of Stinespring dilations supplies general non-sharp bounds [5].

The present contribution is deliberately narrower than a new causal measure. The Ising calculation is a transparent baseline that fixes the subsystem and diamond-norm conventions while extending the known CNOT endpoint to its continuous Cartan axis [15]. The two substantive results are the arbitrary-dimensional partial-SWAP reduction and the complete XY/iSWAP phase diagram. Partial SWAP is used in homogenization and collision models [12, 13, 14] and here exhibits the dimension-dependent transition $\sin \phi = (d^2 - 3)/(d^2 - 1)$. The less symmetric XY line has three analytic regimes, an algebraic quartic in the middle, and an iSWAP endpoint that saturates the global two-qubit bound. A systematic search of the directional-signalling, semicausal-channel, collision-model, and two-qubit-gate literatures found no previous evaluation of Eq. (1) for these continuous families. The search scope and attribution boundary are recorded in Appendix A; this remains a literature assessment rather than a theorem.

The unifying phenomenon is not merely the availability of three formulas. As the coupling grows, the best reduced description can move from the identity to a deliberately noisy channel, and the input witnessing the unavoidable error can change at the same time. The partial-SWAP and XY thresholds make this active-set transition exact. In collision-model language, they identify when retaining the incoming visible state ceases to be the best one-step autonomous approximation.

The distinction from operator entanglement is important. Hilbert–Schmidt distances and the linear entropy of a vectorized unitary are useful algebraic diagnostics [8], but they do not directly give operational channel distinguishability. Every result below is stated in diamond norm and is accompanied by both an upper bound and a matching witness. Likewise, tensor-product-structure geometry [10], approximate operator-algebra correction [9], and exact Lindblad model reduction [11] address related questions of subsystem structure, recoverability, and reduced description, but optimize different objects and do not yield Eq. (1). Operational quantum mereology also seeks dynamically preferred, minimally scrambled subsystems [16], but uses short-time scrambling diagnostics rather than worst-case diamond-norm channel simulation.

Operational picture

Informally, A is a visible system and B is hidden. They interact, B is discarded, and one tries to replace the joint evolution by the best channel acting on A alone. The number $\delta_A(U)$ is the unavoidable worst-case error of that replacement, including inputs correlated with B and with

an external reference. A zero value means that A admits autonomous reduced dynamics. A nonzero value quantifies how strongly information hidden in B can affect the visible output.

Figure 1 shows the most intricate example below. The formulas change because both the optimal effective channel and the input witnessing its error change with interaction strength.

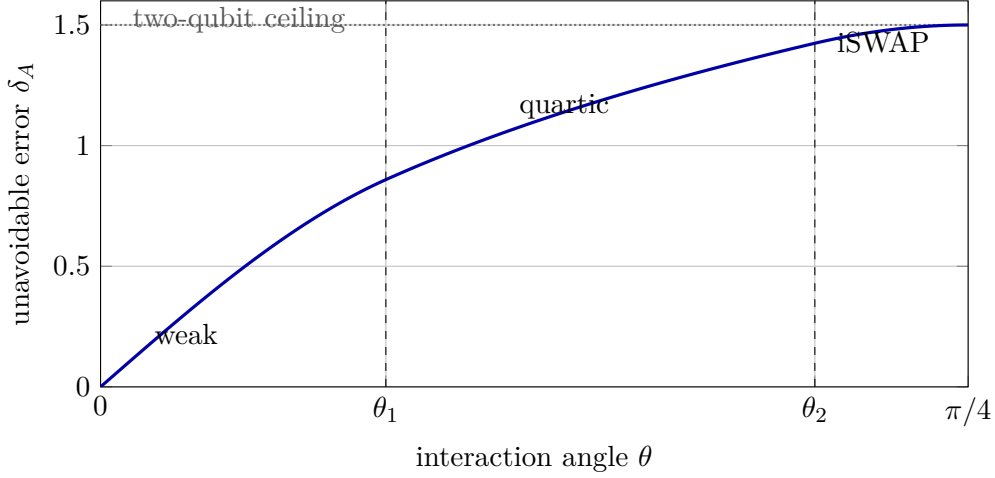


Figure 1: The exact XY directional-signalling curve. The middle segment is the unique physical root of Eq. (32).

2 Preliminaries

Let $\mathcal{L}(H)$ denote the operators on a finite-dimensional Hilbert space H . For a unitary U , write $\text{Ad}_U(X) = UXU^\dagger$. We use the unnormalized Choi operator

$$J(\Phi) = \sum_{j,k} \Phi(|j\rangle\langle k|) \otimes |j\rangle\langle k|.$$

For Hermiticity-preserving $\Phi : \mathcal{L}(X) \rightarrow \mathcal{L}(Y)$, we use the Watrous semidefinite-program characterization [6, 7]. In the unnormalized-Choi convention used here, its Hermiticity-preserving linear primal specialization is

$$\|\Phi\|_\diamond = \max \left\{ \text{Tr}[J(\Phi)Y] : \begin{array}{l} \rho \geq 0, \quad \text{Tr} \rho = 1, \quad Y = Y^\dagger, \\ -I_Y \otimes \rho \leq Y \leq I_Y \otimes \rho \end{array} \right\}. \quad (3)$$

Optimizing first over Y gives the equivalent trace-norm expression

$$\|\Phi\|_\diamond = \max_{\rho \in \mathcal{D}(X)} \|(I_Y \otimes \sqrt{\rho})J(\Phi)(I_Y \otimes \sqrt{\rho})\|_1. \quad (4)$$

The maximizing ρ specifies a physical witness through any purification. If Φ is covariant under input R_g and output S_g , then

$$(\rho, Y) \longmapsto (\overline{R_g} \rho R_g^\top, (S_g \otimes \overline{R_g}) Y (S_g \otimes \overline{R_g})^\dagger)$$

preserves every constraint and the objective in Eq. (3). Haar averaging this complete feasible pair therefore produces an invariant optimal density. This linear averaging, not averaging ρ inside the nonlinear right-hand side of Eq. (4), is the symmetry argument used below. The same SDP also shows that its optimal value is concave in ρ : feasible pairs at two densities can be convexly combined.

We use the maps

$$\mathcal{T} = \text{Tr}_B, \quad \mathcal{N}_U = \text{Tr}_B \circ \text{Ad}_U,$$

so that $\delta_A(U) = \inf_{\mathcal{E}} \|\mathcal{N}_U - \mathcal{E}\mathcal{T}\|_{\diamond}$. Pre- and post-composition by unitary channels are diamond-norm isometries.

Lemma 1 (Covariant twirling). *Suppose a compact group G has unitary representations H_g on $A \otimes B$ and G_g on A such that*

$$\mathcal{N}_U \text{Ad}_{H_g} = \text{Ad}_{G_g} \mathcal{N}_U, \quad \mathcal{T} \text{Ad}_{H_g} = \text{Ad}_{G_g} \mathcal{T}.$$

Then some optimal effective channel in Eq. (1) is covariant under G_g .

Proof. For a candidate $\mathcal{E}$, define $\mathcal{E}_g = \text{Ad}_{G_g^\dagger} \mathcal{E} \text{Ad}_{G_g}$. Covariance gives

$$\mathcal{N}_U - \mathcal{E}_g \mathcal{T} = \text{Ad}_{G_g^\dagger} (\mathcal{N}_U - \mathcal{E} \mathcal{T}) \text{Ad}_{H_g},$$

so every $\mathcal{E}_g$ has the same defect as $\mathcal{E}$. The Haar average $\bar{\mathcal{E}} = \int_G \mathcal{E}_g dg$ is CPTP and covariant. Convexity of the diamond norm implies that its defect is no larger. $\square$

3 The Ising family

Let

$$U_\theta = e^{-i\theta Z \otimes Z} = cI - is Z \otimes Z, \quad c = \cos \theta, \quad s = \sin \theta.$$

Theorem 1 (Exact Ising signalling). *For two qubits,*

$$\delta_A(U_\theta) = |\sin 2\theta| = 2|cs|. \quad (5)$$

An optimal effective channel is

$$\mathcal{E}_\theta(X) = c^2 X + s^2 Z X Z.$$

Proof. Define

$$M(X) = \text{Tr}_B[(I \otimes Z)X], \quad L(X) = -\frac{i}{2}[Z, M(X)].$$

Expanding $U_\theta X U_\theta^\dagger$ and using cyclicity inside the partial trace gives

$$\mathcal{N}_{U_\theta} - \mathcal{E}_\theta \mathcal{T} = 2cs L. \quad (6)$$

The completely bounded trace norm of M is one: its adjoint is the complete operator-norm isometry $Y \mapsto Y \otimes Z$. Also $K \mapsto -i[Z, K]/2$ has completely bounded trace norm at most one by the triangle inequality for left and right unitary multiplication. Hence $\|L\|_{\diamond} \leq 1$, which proves the upper bound $2|cs|$.

For the lower bound, consider

$$\rho_0 = |+\rangle\langle+|_A \otimes |0\rangle\langle 0|_B, \quad \rho_1 = |+\rangle\langle+|_A \otimes |1\rangle\langle 1|_B.$$

They have the same A marginal, so every candidate $\mathcal{E}$ assigns them the same simulated output. Their true reduced outputs differ by $2cs Y$, whose trace norm is $4|cs|$. The triangle inequality therefore forces at least one of the two simulation errors to be at least $2|cs|$. This matches the upper bound. $\square$

At $\theta = \pi/4$, U_θ is locally equivalent to CZ and CNOT. Equation (5) then reproduces the exact CNOT value one, already obtained in Ref. [2]. The continuous formula, rather than that endpoint, is the result of Theorem 1.

4 The partial-SWAP family in dimension d

Let $A \cong B \cong \mathbb{C}^d$ and consider

$$U_\phi = cI - is \text{SWAP}, \quad c = \cos \phi, \quad s = \sin \phi, \quad (7)$$

for $0 \leq \phi \leq \pi/2$. For qubits, the equal-angle Cartan interaction is this family up to global phase [15]:

$$e^{-it(XX+YY+ZZ)} = e^{it} e^{-2it \text{SWAP}}, \quad \phi = 2t.$$

4.1 Reduction to depolarizing channels

The unitary U_ϕ commutes with $V \otimes V$ for every $V \in U(d)$. Lemma 1 therefore permits an optimal effective channel to be chosen covariant under all unitary conjugations. Every such channel is depolarizing:

$$\mathcal{E}_\lambda(X) = \lambda X + (1 - \lambda) \text{Tr}(X) \frac{I}{d}, \quad -\frac{1}{d^2 - 1} \leq \lambda \leq 1. \quad (8)$$

This interval is precisely the CPTP range. Direct expansion gives

$$\mathcal{N}_{U_\phi}(X) = c^2 \text{Tr}_B X + s^2 \text{Tr}_A X - ics \text{Tr}_B(SX - XS),$$

where $\text{Tr}_A X$ is relabelled onto the output copy of A .

4.2 The fixed-channel diamond norm

Define

$$\begin{aligned} r &= 1 - \lambda, & b &= r^2 + d^2 \lambda s^2, \\ h_0 &= \frac{d(d^2 - 3)}{d^2 - 1}, & h_1 &= \frac{2}{d^2 - 1}. \end{aligned} \quad (9)$$

For $b > 0$, also put

$$q = r^2(h_0^2 - h_1^2) + 4b, \quad k = 1 - \frac{r^2 h_1^2}{4b}, \quad z = \frac{r h_0 + \sqrt{q}}{k}. \quad (10)$$

Lemma 2 (Fixed depolarizing channel in dimension d). *Let*

$$d_{d,\phi}(\lambda) = \left\| \mathcal{N}_{U_\phi} - \mathcal{E}_\lambda \mathcal{T} \right\|_\diamond.$$

Apart from the identity point $b = 0$, where the norm vanishes,

$$d_{d,\phi}(\lambda) = \begin{cases} z/d, & r h_1 z \leq 4b, \\ 2r(h_0 + h_1)/d, & r h_1 z > 4b. \end{cases} \quad (11)$$

The quantities b and k are positive and q is nonnegative on the CPTP interval away from that identity point, so the closed form is well defined. The lemma is asserted, and proved, on the whole admissible interval $-1/(d^2 - 1) \leq \lambda \leq 1$, including negative λ ; the outer minimization later uses only $0 \leq \lambda \leq 1$, so the closed form is established on a strictly larger range than is subsequently needed.

Proof. The Watrous SDP is invariant under the simultaneous $V \otimes V$ covariance action. We average the full linear primal feasible tuple, not the nonlinear right-hand side of Eq. (4). Affineness preserves feasibility and objective value, and Schur–Weyl duality then permits an optimal input density

$$\rho = xP_- + yP_+, \quad r_-x + r_+y = 1, \quad r_{\pm} = \frac{d(d \pm 1)}{2}.$$

Let $u = r_+y$ be its total symmetric-subspace weight. Put

$$\kappa_+ = \frac{(d-1)(d+2)}{d+1}, \quad \kappa_- = \frac{(d+1)(d-2)}{d-1}, \quad \mathcal{B} = (1+\lambda)^2 - 4\lambda c^2.$$

For

$$K_\rho = (I \otimes \sqrt{\rho})J(\mathcal{N}_{U_\phi} - \mathcal{E}_\lambda \mathcal{T})(I \otimes \sqrt{\rho}),$$

the block decomposition can be made explicit. On output–input space $A_{\text{out}} \otimes A_{\text{in}} \otimes B_{\text{in}}$ define

$$|v_a^\pm\rangle = \frac{1}{\sqrt{2(d \pm 1)}} \sum_{j=1}^d |j\rangle \otimes (|j, a\rangle \pm |a, j\rangle), \quad a = 1, \dots, d.$$

The vectors $v_a^\pm$ span the only subspaces coupled between the symmetric and antisymmetric input sectors. Inside $A_{\text{out}} \otimes \text{Sym}^2(\mathbb{C}^d)$, the orthogonal complement of $\text{span}\{v_a^+\}$ carries the scalar $-(1-\lambda)y/d$. Inside $A_{\text{out}} \otimes \wedge^2(\mathbb{C}^d)$, the orthogonal complement of $\text{span}\{v_a^-\}$ carries $-(1-\lambda)x/d$. For each a , the two-dimensional space $\text{span}\{v_a^-, v_a^+\}$ carries, in that order, the block

$$\begin{pmatrix} \frac{(d+1)(d-2)}{2d}(1-\lambda)x & \bar{\omega} \\ \omega & \frac{(d-1)(d+2)}{2d}(1-\lambda)y \end{pmatrix}, \quad |\omega|^2 = \frac{d^2-1}{4}xy\mathcal{B}.$$

The phase of ω depends on the phase convention for $v_a^\pm$ and does not affect the spectrum. Using $u = r_+y$, $1-u = r_-x$, its trace and determinant give the remaining roots below. Thus the complete spectrum is

eigenvalue	multiplicity
$-(1-\lambda)y/d$	$d(d-1)(d+2)/2$
$-(1-\lambda)x/d$	$d(d+1)(d-2)/2$
ν_+, ν_-	d each

where

$$\begin{aligned} \nu_+ + \nu_- &= \frac{1-\lambda}{d^2}[\kappa_+u + \kappa_-(1-u)], \\ \nu_+\nu_- &= \frac{u(1-u)}{d^2} \left[\frac{(1-\lambda)^2}{d^2}\kappa_+\kappa_- - \mathcal{B} \right]. \end{aligned} \tag{12}$$

Indeed $\kappa_+\kappa_- = d^2 - 4$ and

$$\mathcal{B} - \frac{(1-\lambda)^2}{d^2}\kappa_+\kappa_- = \frac{4b}{d^2} \geq 0,$$

so the last product is nonpositive even when $\lambda < 0$. Summing all eigenvalue magnitudes gives

$$d_{d,\phi}(\lambda) = \max_{0 \leq u \leq 1} \left\{ \frac{1-\lambda}{d}[\kappa_+u + \kappa_-(1-u)] + \sqrt{\frac{(1-\lambda)^2}{d^2}[\kappa_+u - \kappa_-(1-u)]^2 + 4u(1-u)\mathcal{B}} \right\}. \tag{13}$$

With $t = 2u - 1$, elementary algebra turns Eq. (13) into

$$\max_{-1 \leq t \leq 1} \frac{1}{d} \left[r(h_0 + h_1 t) + \sqrt{r^2(h_0 + h_1 t)^2 + 4b(1 - t^2)} \right]. \quad (14)$$

Here one uses $\kappa_+ \kappa_- = d^2 - 4 = h_0^2 - h_1^2$; this identity accounts for the apparently different linear and radical coefficients in Eqs. (13) and (14). If the maximum is interior, its location is $t_* = rh_1 z / (4b)$ and its value is z/d . Since every factor in t_* is nonnegative, a maximum at $t = -1$ cannot occur. The condition $t_* \leq 1$ gives the first branch of Eq. (11); otherwise the maximum is at $t = 1$ and gives its second branch. The dimensions of the two scalar complements are respectively $d(d-1)(d+2)/2$ and $d(d+1)(d-2)/2$, so the displayed blocks account for all d^3 eigenvalues. The maximizing ρ , together with any purification, realizes the matching lower-bound witness; Eq. (3) supplies the upper bound. $\square$

Remark 1 (Status of the general- d spectrum). The block decomposition just given is a hand calculation valid for every $d \geq 2$: the vectors $v_a^\pm$ are defined uniformly in d , and the stated multiplicities $d(d-1)(d+2)/2$, $d(d+1)(d-2)/2$, and d , d sum to d^3 , so no eigenvalue is unaccounted for at any dimension. The accompanying script `tools/verify_block_reductions.py` reconstructs the unreduced physical Choi operator and confirms this spectrum for $d = 2, 3, 4$ only. Those checks are therefore a spot check of the algebra above, not the source of the all- d claim, which rests on the Schur–Weyl argument in the proof.

4.3 Optimization and the dimension-dependent threshold

Theorem 2 (Partial-SWAP signalling in arbitrary dimension). *For every $d \geq 2$,*

$$\delta_A(U_\phi) = \min_{-1/(d^2-1) \leq \lambda \leq 1} d_{d,\phi}(\lambda), \quad (15)$$

with $d_{d,\phi}$ given explicitly by Eq. (11). Moreover,

$$0 \leq \sin \phi \leq \frac{d^2 - 3}{d^2 - 1} \implies \lambda_* = 1, \quad \delta_A(U_\phi) = 2 \sin \phi. \quad (16)$$

For stronger non-SWAP coupling, a minimizer lies in $0 < \lambda_* < 1$. Equation (15) is an exact one-dimensional convex characterization in this regime. For qubits the interior minimizer is algebraic: writing $s = \sin \phi$, it satisfies the unsquared stationarity equation

$$3(\lambda_* - 1 + 2s^2) \left[\sqrt{(1 + \lambda_*)^2 - 4\lambda_*(1 - s^2)} - \frac{2(1 - \lambda_*)}{3} \right] = (1 + \lambda_*)^2 - 4\lambda_*(1 - s^2). \quad (17)$$

In particular it is a root of the explicit quartic

$$\begin{aligned} \mathcal{P}(\lambda, s) = & 2\lambda^4 + (18s^2 - 8)\lambda^3 + (45s^4 - 43s^2 + 12)\lambda^2 \\ & + (36s^6 - 54s^4 + 32s^2 - 8)\lambda + 5s^4 - 7s^2 + 2, \quad \mathcal{P}(\lambda_*, s) = 0. \end{aligned} \quad (18)$$

Roots introduced by eliminating the square root are discarded using Eq. (17). In particular, positivity of its right-hand side and of the bracket on the physical branch forces

$$\lambda_* > 1 - 2s^2. \quad (19)$$

The quartic may have an additional root in $(0, 1)$, but that root solves the conjugate equation with the opposite square-root sign. The endpoint factorizations

$$\mathcal{P}(0, s) = (s^2 - 1)(5s^2 - 2), \quad \mathcal{P}(1, s) = 4s^4(9s^2 - 1)$$

Table 1: Dimension dependence of the partial-SWAP weak threshold and the known full-SWAP endpoint.

d	weak threshold for $\sin \phi$	$\delta_A(S)$
2	1/3	3/2
3	3/4	16/9
4	13/15	15/8
d	$(d^2 - 3)/(d^2 - 1)$	$2(1 - 1/d^2)$

for the left-hand side $\mathcal{P}$ of Eq. (18) make these possible extraneous-root changes explicit. For general d , Eq. (15) remains the exact characterization; we do not claim a useful generic radicals expression after the outer minimization. At $\phi = \pi/2$,

$$\lambda_* = 0, \quad \delta_A(S) = 2 \left(1 - \frac{1}{d^2} \right). \quad (20)$$

Proof. Equation (15) follows from covariance and Lemma 2. The function $d_{d,\phi}(\lambda)$ is convex because it is the diamond norm of a map affine in λ . At $\lambda = 1$,

$$d_{d,\phi}(1) = 2s, \quad \left. \frac{d}{d\lambda} d_{d,\phi}(\lambda) \right|_{1-} = s - \frac{d^2 - 3}{d^2 - 1}.$$

Convexity proves Eq. (16). The matching invariant density has equal total symmetric and antisymmetric weights,

$$\rho = \frac{dI - S}{d(d^2 - 1)}.$$

At $\lambda = 0$ and $c > 0$, the derivative is $-2(d^2 - 1)c^2/d^2 < 0$. Convexity excludes negative λ from the outer optimum, while the positive derivative at $\lambda = 1$ in the strong regime places a minimizer in $(0, 1)$. At SWAP the minimizer is $\lambda = 0$; Eq. (14) is maximized at $t = 1/d$, corresponding to $\rho = I/d^2$, and yields Eq. (20), in agreement with Ref. [2].

For $d = 2$ and $1/3 < s < 1$, differentiating the interior-maximizer branch of Eq. (11) gives Eq. (17). On $0 < \lambda < 1$ the interior-maximizer branch is automatic, since

$$b - \frac{4(1 - \lambda)^2}{9} = \frac{5(1 - \lambda)^2 + 36\lambda s^2}{9} > 0.$$

Squaring once gives $-4\mathcal{P}(\lambda, s) = 0$. Equation (19) records the sign lost in this squaring step. Conversely, any $\lambda \in (0, 1)$ satisfying the unsquared equation has zero derivative and is therefore globally minimizing by convexity. There can be only one such root: two distinct stationary minimizers would make the convex analytic objective constant on an interval, contrary to its nonconstant derivative. Thus Eq. (19) selects λ_* from any conjugate-sign quartic root. $\square$

5 The XY/iSWAP line

Consider

$$U_\theta = e^{-i\theta(XX+YY)}, \quad 0 \leq \theta \leq \frac{\pi}{4}, \quad C = \cos 2\theta, \quad S = \sin 2\theta. \quad (21)$$

The zero- and two-excitation states are fixed, while on $\text{span}\{|01\rangle, |10\rangle\}$ the unitary is $CI - iSX$. Equivalently,

$$U_\theta = \cos^2 \theta II - \frac{i}{2} \sin 2\theta (XX + YY) + \sin^2 \theta ZZ.$$

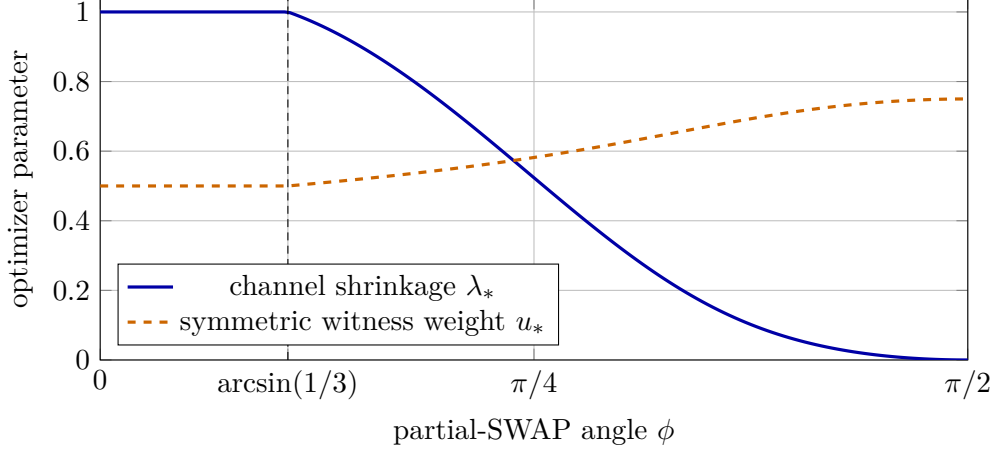


Figure 2: Optimizer transition for qubit partial SWAP. The effective channel remains the identity until $\sin \phi = 1/3$ and then becomes progressively more depolarizing. Simultaneously, the invariant diamond witness shifts from equal total symmetric/antisymmetric weight $u_* = 1/2$ toward $u_* = 3/4$ at SWAP. The plotted interior values solve the scalar optimization in Eq. (15).

5.1 Axial covariance and the fixed-channel norm

Joint Pauli covariance first restricts an optimal effective channel to a Pauli channel. The additional axial symmetry generated by $ZI + IZ$ makes its X and Y weights equal. We therefore write

$$\mathcal{E}_{u,w} : (r_x, r_y, r_z) \mapsto (ur_x, ur_y, wr_z), \quad -1 \leq w \leq 1, \quad |u| \leq \frac{1+w}{2}. \quad (22)$$

The corresponding Pauli probabilities are

$$q_I = \frac{1+2u+w}{4}, \quad q_X = q_Y = \frac{1-w}{4}, \quad q_Z = \frac{1+w-2u}{4}.$$

Lemma 3 (XY fixed-channel norm). *For fixed (u, w) in Eq. (22), put $B = \frac{1}{2} - A$ and*

$$\alpha = \frac{A(1-w)}{2}, \quad \gamma^2 = \frac{AB}{2}, \quad g = C - u, \quad h = S, \quad \beta = \frac{B}{2} [w - (C^2 - S^2) + 2iSC].$$

Define

$$M_3(A) = \begin{pmatrix} \alpha & \gamma(g+ih) & \gamma(-g+ih) \\ \gamma(g-ih) & 0 & \beta \\ \gamma(-g-ih) & \bar{\beta} & 0 \end{pmatrix}. \quad (23)$$

Then

$$\|\mathcal{N}_{U_\theta} - \mathcal{E}_{u,w}\mathcal{T}\|_\diamond = \max_{0 \leq A \leq 1/2} 2[\alpha + \|M_3(A)\|_1]. \quad (24)$$

Proof. Average the complete linear primal diamond-SDP tuple over joint axial rotations and $X \otimes X$; averaging only the density in Eq. (4) would not justify the reduction. The residual odd-sector coherence changes sign under the real-linear involution

$$X \mapsto (Z \otimes I)X^*(Z \otimes I),$$

which is a covariance because $U_\theta(Z \otimes I) = (Z \otimes I)U_\theta^*$. Averaging this involution leaves an optimal density of the form

$$\rho = A(|00\rangle\langle 00| + |11\rangle\langle 11|) + B(|01\rangle\langle 01| + |10\rangle\langle 10|).$$

Inserting ρ in Eq. (4), the scaled Choi operator commutes with the axial charge

$$Q_{\text{ax}} = \frac{1}{2}(Z_{A_{\text{out}}} - Z_{A_{\text{in}}} - Z_{B_{\text{in}}}),$$

where the input signs reflect the conjugate covariance action. Its four charge sectors have dimensions 1, 3, 3, 1. The sector spanned by $|011\rangle$ contributes the eigenvalue $-\alpha$, while the ordered basis

$$|000\rangle, \quad \frac{|101\rangle + |110\rangle}{\sqrt{2}}, \quad \frac{|101\rangle - |110\rangle}{\sqrt{2}}$$

gives Eq. (23). The symmetry $X^{\otimes 3}$ reverses the charge and carries these sectors to the isolated vector $|100\rangle$ and the three-dimensional sector formed from $|111\rangle, |010\rangle, |001\rangle$. The latter is unitarily equivalent after conjugating phases. This accounts for all eight dimensions and proves Eq. (24). For reference, the characteristic polynomial $x^3 + p_2x^2 + p_1x + p_0$ of M_3 has

$$\begin{aligned} p_2 &= -\alpha, \\ p_1 &= -\left[2\gamma^2(g^2 + h^2) + |\beta|^2\right], \\ p_0 &= \alpha|\beta|^2 + 2\gamma^2(g^2 - h^2)\Re\beta - 4\gamma^2gh\Im\beta. \end{aligned}$$

□

Lemma 3 is an exact finite-dimensional answer along the whole XY line:

$$\delta_A(U_\theta) = \min_{\substack{-1 \leq w \leq 1 \\ |u| \leq (1+w)/2}} \max_{0 \leq A \leq 1/2} 2[\alpha + \|M_3(A)\|_1]. \quad (25)$$

Whenever M_3 has one nonnegative and two nonpositive eigenvalues, $\text{Tr } M_3 = \alpha$ gives $2[\alpha + \|M_3\|_1] = 4\lambda_{\max}(M_3)$. Two coupling regimes simplify further.

5.2 Closed weak and iSWAP branches

Let $\theta_1 = 0.258270520262\dots$ be the unique root in $(0, \pi/4)$ of

$$2SC = 3C^3 + C^2 - C - 1, \quad (26)$$

and let $\theta_2 = 0.646615513406\dots$ solve

$$1 + C^2 = 2C(1 + S). \quad (27)$$

Equivalently, $\tan \theta_2$ is the positive root of $t^3 + t^2 - 1 = 0$. The middle interval lies inside $(\pi/24, 5\pi/24)$. Indeed, direct substitution at $\theta = \pi/24$ gives

$$2SC - (3C^3 + C^2 - C - 1) = 1 - \frac{11\sqrt{2} + 5\sqrt{6}}{16} - \frac{\sqrt{3}}{4} < 0,$$

so $\theta_1 > \pi/24$. The polynomial $t^3 + t^2 - 1$ is strictly increasing for $t > 0$ and at $t = \tan(5\pi/24)$ equals $-96 - 67\sqrt{2} + 55\sqrt{3} + 39\sqrt{6} > 0$, so $\theta_2 < 5\pi/24$. Consequently

$$2SC = \sin 4\theta > \frac{1}{2} \quad \text{throughout } (\theta_1, \theta_2). \quad (28)$$

Theorem 3 (Closed XY signalling branches). *For $0 \leq \theta \leq \theta_1$,*

$$\delta_A(U_\theta) = \sin 4\theta, \quad (q_I, q_X, q_Y, q_Z) = \left(C^2, \frac{S^2}{2}, \frac{S^2}{2}, 0\right). \quad (29)$$

For $\theta_2 \leq \theta \leq \pi/4$,

$$\delta_A(U_\theta) = S + \frac{S^2}{2}, \quad (30)$$

with

$$(q_I, q_X, q_Y, q_Z) = \frac{1}{4} \left(1 + C^2 + 2C(1 + S), S^2, S^2, 1 + C^2 - 2C(1 + S) \right). \quad (31)$$

In particular, $\delta_A(\text{iSWAP}) = 3/2$.

Proof. For every effective channel, the choice $A = 0$ in Eq. (24) gives

$$\sqrt{[w - (C^2 - S^2)]^2 + 4S^2C^2} \geq 2SC.$$

This is the weak lower bound. For the channel in Eq. (29), let $K_1 = (SC/2)I - M_3(A)$. Its leading principal minor is $S(C - 2AS)/2$, and its $(2, 3)$ principal minor is $AC^2(A - 1)(C - 1)(C + 1)$. The other two order-two principal minors coincide and equal

$$\left(\frac{SC}{2} - AS^2 \right) \frac{SC}{2} - \frac{A(1/2 - A)}{2} [C^2(1 - C)^2 + S^2].$$

As a quadratic in A , its leading coefficient is $(C - 1)(C^3 - C^2 - C - 1)/2 > 0$. With $\tau = \tan \theta$, its discriminant is

$$\frac{\tau^4}{(1 + \tau^2)^8} \left(\tau^{12} - 10\tau^{10} - 8\tau^9 + 31\tau^8 + 16\tau^7 - 36\tau^6 - 32\tau^5 + 23\tau^4 + 16\tau^3 + 14\tau^2 + 8\tau - 7 \right).$$

The first threshold obeys $\tan \theta_1 < 27/100$; this rational enclosure is established in Appendix B from the strict sign change $p_1(1/4) < 0 < p_1(27/100)$ of the exact threshold polynomial, and it is what makes the following estimate valid across the entire weak interval rather than at sampled angles. Since $0 < \tau < 27/100$ throughout, and since every positive monomial in the bracket is increasing in τ on that range, dropping the negative monomials and evaluating each positive monomial at the *upper* enclosure $\tau = 27/100$ gives an upper bound for the bracket excluding its constant term; that sum is below $3.63 < 7$, so the discriminant is nonpositive. Completing the square therefore proves both coincident minors nonnegative. Finally,

$$\det K_1 = \frac{AC(C - 1)}{4} K(A),$$

where K is convex and

$$K(0) = S[2SC - (3C^3 + C^2 - C - 1)], \quad K(1/2) = -C(C + 1)S(C - S).$$

Both endpoint values are nonpositive precisely through the weak interval; convexity gives $K(A) \leq 0$. Moreover,

$$\det M_3 = -\frac{AC^2(2A - 1)^2(C - 1)^3(C + 1)}{4} \geq 0,$$

while the sum of its order-two principal minors is nonpositive. Thus M_3 has one nonnegative and two nonpositive eigenvalues: three positive eigenvalues would make the order-two sum positive, while two positive and one negative would make the determinant negative; the zero cases follow by continuity. The principal minors above give $K_1 \geq 0$, hence $\lambda_{\max}(M_3) \leq SC/2$; together with the inertia identity following Eq. (25), this makes Eq. (24) at most $2SC$. This proves Eq. (29) and identifies the first threshold.

For the strong branch, Eq. (31) has $u = C(1 + S)$ and $w = C^2$. It is CPTP exactly when Eq. (27) holds with the strong-side inequality. At $A = 1/4$, the largest eigenvalue of M_3 is

$$L = \frac{S(2 + S)}{8},$$

with eigenvector proportional to $(i\sqrt{2}, 1, 1)$. The other two eigenvalues have sum $-S/4$ and product $S^2(7S^2 + 2S - 8)/64$, so they are negative throughout this regime. Indeed $S \geq S(\theta_2) > 0.96$, whereas the positive root of $7S^2 + 2S - 8$ is below 0.94. The derivatives of L in both u and w vanish. Since $\lambda_{\max}$ is convex for an affine Hermitian matrix family, the fixed $A = 1/4$ witness gives the global lower bound $4L = S + S^2/2$.

It remains to prove the matching upper bound. Put $H_3 = LI - M_3(A)$. Direct factorization gives

$$\det H_3 = -\frac{S^3(4A - 1)^2(S + 2)}{512} \left[A(28S^2 + 8S - 32) + 12 - 13S^2 - 4S \right].$$

The bracket is negative at both endpoints of $A \in [0, 1/2]$. The two nontrivial order-two principal minors are $S^2G_1/64$ and $S^2G_2/64$, where, with $t = 2A$,

$$\begin{aligned} G_1 &= (16 - 8S^2)t^2 + (6S^2 - 4S - 16)t + (S + 2)^2, \\ G_2 &= (12S^2 - 16)t^2 + (32 - 24S^2)t + 13S^2 + 4S - 12. \end{aligned}$$

G_1 has positive completed-square minimum, while G_2 is concave with nonnegative endpoint values $13S^2 + 4S - 12$ and $(S + 2)^2$; the first is already positive for $S > 0.82$, below the strong-regime range. Together with $(H_3)_{11} = S(S + 2 - 4AS)/8$, all principal minors are nonnegative, so $H_3 \geq 0$. In addition,

$$\det M_3 = \frac{AS^3(2A - 1)^2(S + 2)(7S^2 + 2S - 8)}{32} \geq 0,$$

and the sum of its order-two principal minors is nonpositive. Thus again M_3 has one nonnegative and two nonpositive eigenvalues, and Eq. (24) is at most $4L$. This proves Eq. (30). At $S = 1$, the four probabilities are $1/4$ and the value is $3/2$. $\square$

Before treating the intermediate regime we isolate its convex-analytic skeleton, so that the algebraic elimination that follows can be carried out against a single, already-justified optimality certificate.

Lemma 4 (Facet selection and convex KKT sufficiency). *Fix θ and parametrize the axially covariant effective channels of Eq. (22) by their Bloch eigenvalues (u, u, w) . Then the following hold.*

- (i) (Convexity.) *The admissible set $\mathcal{T} = \{(u, w) : -1 \leq w \leq 1, |u| \leq (1 + w)/2\}$ is a triangle with vertices $(0, -1)$, $(1, 1)$, and $(-1, 1)$, and the facet $q_Z = 0$ is the edge $w = 2u - 1$. The fixed-channel diamond norm $F(u, w) = \|\mathcal{N} - \mathcal{E}_{u,w}\mathcal{T}\|_\diamond$ is convex and continuous on $\mathcal{T}$, being the norm of a map affine in (u, w) .*
- (ii) (Concavity and saddle existence.) *For fixed (u, w) the Watrous primal value is concave in the invariant witness parameter $A \in [0, 1/2]$. Consequently the optimization is convex-concave and its min-max value is attained at a saddle. On any parameter interval where both the maximizing witness and the minimizing channel are unique, Berge's maximum theorem makes their coordinates continuous.*
- (iii) (Sufficiency.) *Suppose $(u_\star, w_\star)$ lies in the relative interior of the edge $w = 2u - 1$, the directional derivative of F along that edge vanishes, and F admits a subgradient whose normal component points into $q_Z \geq 0$. Then $(u_\star, w_\star)$ minimizes F over the whole triangle $\mathcal{T}$, not merely over the edge.*

Proof. (i) The vertex description is immediate from the constraints, and $q_Z = 0$ is exactly $w = 2u - 1$. The map $\mathcal{E}_{u,w}$ is affine in (u, w) , hence so is $\mathcal{N} - \mathcal{E}_{u,w}\mathcal{T}$; a norm of an affine function is convex, and convex finite functions on a polytope are continuous.

(ii) The feasible pairs (ρ, Y) of the Watrous primal form a convex set and the objective is linear, so the value is concave in ρ ; the invariant density is affine in A , giving concavity in A . A convex–concave function on a product of compact convex sets has a saddle by Sion’s minimax theorem. Berge’s maximum theorem gives upper hemicontinuity of the optimizer correspondences; when each is single-valued, this is ordinary continuity.

(iii) This is the standard first-order condition for a convex function on a polytope. Vanishing derivative along the edge together with a subgradient whose normal component points inward means that the negative of some subgradient lies in the outward normal cone of $\mathcal{T}$ at (u_*, w_*) , i.e. $0 \in \partial F(u_*, w_*) + N_{\mathcal{T}}(u_*, w_*)$. For a convex F this normal-cone inclusion is sufficient, not merely necessary, for a global minimum over $\mathcal{T}$. In particular no other edge or vertex of the triangle can beat it, so the remaining facets need not be examined separately. $\square$

Lemma 4(iii) is what reduces the intermediate regime to a single algebraic computation: it suffices to exhibit one facet point with vanishing tangential derivative and a positive normal multiplier.

Theorem 4 (Intermediate XY branch). *For $\theta_1 < \theta < \theta_2$, put $t = \tan \theta$. Then $\delta_A(U_\theta)$ is the unique root $\Delta \in (1/2, 2)$ of*

$$\begin{aligned} Q_+(\Delta, t) = & (t^6 + t^5 + 3t^4 + 2t^3 + 3t^2 + t + 1)\Delta^4 \\ & + (2t^7 - 46t^6 - 122t^5 - 170t^4 - 138t^3 - 58t^2 - 14t + 2)\Delta^3 \\ & + (-84t^7 + 24t^6 + 72t^5 + 96t^4 - 68t^3 - 56t^2 - 32t)\Delta^2 \\ & + (96t^7 + 64t^6 + 160t^5 + 384t^4 + 320t^3 + 128t^2)\Delta \\ & + 64t^5(2t^2 + 2t + 1). \end{aligned} \quad (32)$$

Remark 2 (Computer-assisted status). Theorem 4 is a computer-assisted result. The convex analysis selecting the active facet, recorded in Lemma 4 and in the first part of the proof below, is fully human-checkable. The subsequent eliminations are not: the exclusion of the conjugate sheet W_- , the resultant factorization producing Q_+ , the positivity of its cofactors, and the Sturm root count are polynomial identities and sign conditions in up to four variables with integer coefficients of large height. They are stated in Appendix B and *verified in exact rational arithmetic* by the script `tools/derive_xy_middle.py` accompanying this manuscript, which is part of the proof of record rather than an illustration. Because every such step is an identity or a strict inequality between polynomials with exact rational coefficients, the verification is decidable and involves no floating-point arithmetic, rounding, or numerical root finding. A reader wishing to check Theorem 4 independently should re-run that script, or re-derive the same resultants and Sturm sequences in any computer algebra system.

Proof. We first prove the active facet. On $qz = 0$, write $w = 2u - 1$, $B = 1/2 - A$, and let ℓ be the largest eigenvalue of $M_3(A)$. Put

$$m = (u - C)^2 + S^2, \quad N = (u - C^2)^2 + C^2 S^2, \quad \Theta = u(1 - u)(1 - C)^2.$$

The characteristic polynomial becomes

$$P(\ell; A, u) = \ell^3 - A(1 - u)\ell^2 - (ABm + B^2N)\ell - AB^2\Theta. \quad (33)$$

Thus $\det M_3 = AB^2\Theta \geq 0$, while the sum of its order-two principal minors is $-(ABm + B^2N) \leq 0$. In the interior M_3 has one positive and two negative eigenvalues, so the fixed-witness objective is 4ℓ and $P_\ell > 0$.

At $A = 0$, the largest root is $\ell_0 = \sqrt{N}/2$. Implicit differentiation of Eq. (33), using $P_\ell(\ell_0; 0, u) = N/2$, gives

$$4\ell'(0) = \frac{2}{N}\Xi(u), \quad \Xi(u) = (1 - u)N + \Theta - \sqrt{N}(2N - m). \quad (34)$$

Thus Ξ is the exact one-sided activity test for moving from $A = 0$ into $A > 0$.

For fixed channel parameters, the diamond-SDP value is concave in its input density, and the resulting facet objective $f(u)$ is convex and continuous, by Lemma 4(i)–(ii). In the interior the maximizing A is unique: concavity makes its maximizer set an interval, while a nontrivial interval would make the cubic $P(\ell; A, u)$ vanish identically in A despite its nonzero leading coefficient $-\Theta$. If its unique maximizing witness were $A = 0$, Danskin's theorem would give

$$f'(u) = \frac{2(u - C^2)}{\sqrt{(u - C^2)^2 + C^2 S^2}},$$

so an interior channel minimizer could occur only at $u = C^2$. At that point Eq. (34) reduces to

$$\Xi(C^2) = C(1 - C)S[2SC - (3C^3 + C^2 - C - 1)],$$

which is positive after θ_1 ; hence $A = 0$ is not maximizing there. The endpoint derivatives in u point inward, while the derivative toward smaller A excludes $A = 1/2$. Thus the middle saddle has $0 < u < 1$ and $0 < A < 1/2$.

A flat interval of minimizing u values is impossible. Indeed,

$$\text{disc}_A P = \frac{(C - 1)^2 \ell^2 (2\ell + u)}{16} \mathcal{R}(u, \ell, C), \quad [u^7] \mathcal{R} = (1 - C)^2 - 8\ell.$$

If $A = 0$ on a subinterval, the boundary value $2\sqrt{(u - C^2)^2 + C^2 S^2}$ is strictly convex and cannot be constant. Hence any flat interval contains an open subinterval with $0 < A < 1/2$, where the displayed discriminant applies. At every point of that subinterval, witness stationarity gives $P = P_A = 0$, so eliminating A makes $\text{disc}_A P$ vanish at the same fixed value of ℓ . Constancy on an interval would therefore force $\mathcal{R}(u, \ell, C)$ to vanish identically as a polynomial in u , hence $4\ell = (1 - C)^2/2 < 1/2$, whereas the $A = 0$ witness gives $4\ell \geq 2SC > 1/2$ by Eq. (28). Both optimizer coordinates are therefore unique, and Lemma 4(ii) makes the facet saddle continuous in θ .

It remains to check the normal KKT multiplier. Holding w fixed before restricting to the facet gives

$$P_u = 2ABG, \quad G = (C - u)\ell + B\Psi, \quad \Psi = u^2 - C(1 + C)u + C.$$

Here $\Psi > 0$ because its discriminant is $C(C - 1)(C^2 + 3C + 4) < 0$. Eliminating A, u, ℓ from $P = P_A = P_u + 2P_w = G = 0$, after saturating the interior factors, gives

$$5C^3 + C^2 + 3C - 1 = 0.$$

This is exactly the second-threshold equation. At θ_1 the limiting weak saddle has $A = 0$, $u = C^2$, $\ell = SC/2$, and

$$G = \frac{C(1 - C)}{2}[(1 + C) + SC] > 0.$$

Therefore $G > 0$ throughout (θ_1, θ_2) : by the continuity just proved, the facet saddle varies continuously in θ , so G can change sign on the open interval only by vanishing there, and the elimination above shows that this happens only at θ_2 . The gradient of the convex channel objective is then a positive multiple of the inward normal to $q_Z \geq 0$: tangential stationarity gives zero directional derivative along the facet, while $G > 0$ supplies a subgradient with inward normal component, whose negative lies in the outward normal cone. Lemma 4(iii) applies verbatim, so the facet point is the global minimizer over the entire CPTP triangle, and no other edge or vertex needs to be examined. Consequently $\Delta = 4\ell = \delta_A(U_\theta)$. The $A = 0$ witness and Eq. (28) give $\Delta \geq 2SC > 1/2$, while Eq. (2) gives $\Delta \leq 3/2 < 2$. This is the root window used below.

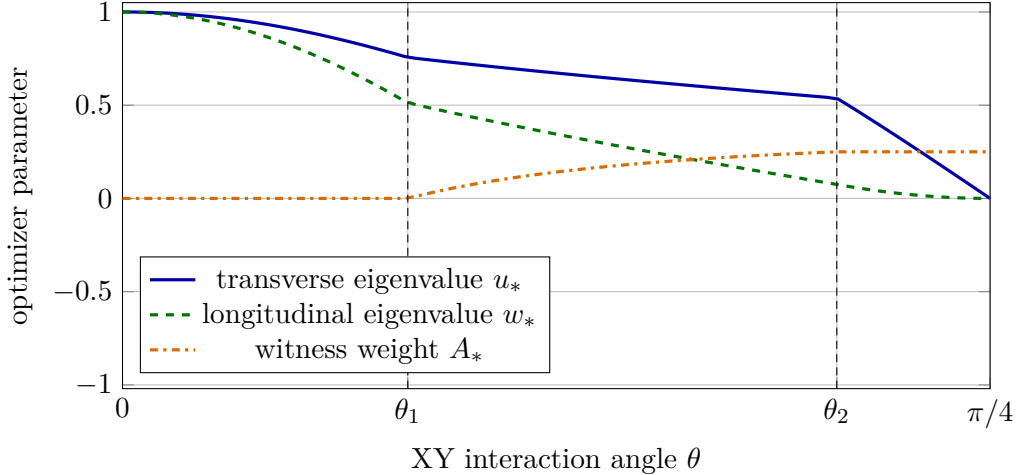


Figure 3: Optimal effective-channel eigenvalues and invariant witness along the XY line. The weak regime has $A_* = 0$; the middle saddle moves along the CPTP facet $w_* = 2u_* - 1$ with $0 < A_* < 1/2$; and the strong regime locks to $A_* = 1/4$. The middle values are a numerical reconstruction of the unique algebraic saddle proved in Theorem 4, included for visualization rather than as proof.

It remains to eliminate the unique saddle and identify the physical algebraic sheet. After putting $a = 2A$ and $\Delta = 4\ell$, the stationary characteristic polynomial is quadratic in u . Its discriminant splits into two conjugate factors $W_+(a, \Delta, \pm t)$. An exact rational lower bound excludes the minus sheet throughout an enclosing rational window for the middle interval. Eliminating a from the remaining sheet and witness stationarity gives $Q_+(\Delta, t)$ times factors that are strictly positive on the physical domain. Finally, the discriminant of Q_+ is nonzero on that window, and an exact Sturm count gives one root in $(1/2, 2)$. Substitution of the two closed branches proves exact joining at both thresholds. All factors, interval bounds, and saturation steps are given in Appendix B; no floating-point root selection enters this argument. As recorded in Remark 2, these four eliminations are carried out and checked in exact rational arithmetic by the accompanying script rather than by hand.

Finally, we record in what sense this value is certified from below. In the weak and strong regimes the lower bound is an explicitly exhibited input state. Here, by contrast, the certificate is the convex-concave saddle of Lemma 4(ii): the value is attained at the interior invariant witness $\rho(A_*)$ of the unique saddle, and Sion's minimax theorem guarantees that no effective channel can do better against it. The witness is therefore operational and matching, but it is specified algebraically as the saddle coordinate rather than in closed form. $\square$

The full curve satisfies $\delta_A(\theta) = \delta_A(\pi/2 - \theta)$ and returns to zero at $\theta = \pi/2$, where $U_{\pi/2} = Z \otimes Z$. The symmetry follows from $U_{\pi/2 - \theta} = (Z \otimes Z) \overline{U_\theta}$, together with invariance of the diamond norm under unitary conjugation and transposition.

6 Discussion

The Ising family provides the baseline mechanism: one dephasing channel is optimal for every coupling, and two product inputs already witness the exact defect. The main transitions occur in the other two families. Along partial SWAP, the optimal effective channel itself changes: it remains the identity until $\sin \phi = (d^2 - 3)/(d^2 - 1)$, then moves into the depolarizing interval and reaches complete depolarization at SWAP. The optimal diamond witness likewise generally uses a reference system, as encoded by a purification of the symmetric/antisymmetric density in

Lemma 2; Fig. 2 displays both optimizer trajectories for qubits.

The XY line has less symmetry and a genuine active-set phase diagram. Its weak branch is controlled by the one-excitation witness, its central iSWAP branch by the maximally mixed witness, and the intervening interval moves through an algebraic saddle, as shown in Fig. 3. The iSWAP value $3/2$ equals the qubit SWAP value despite the gates belonging to different local-equivalence classes in the canonical classification of two-qubit gates [15]. By the universal bound (2), which applies to every unitary on $\mathbb{C}^2 \otimes \mathbb{C}^2$, iSWAP is therefore a maximally signalling two-qubit unitary, not merely a gate that happens to share SWAP’s value.

The threshold is operational rather than spectral: it is where the outer channel optimization first benefits from adding depolarization. It is not visible from operator entanglement alone. This supplies a concrete example in which a coarse-grained autonomy question differs from a Hilbert–Schmidt interaction diagnostic. For repeated-interaction and homogenization models, the threshold also separates a regime in which the incoming visible state is the best autonomous predictor from one in which deliberately forgetting part of that state improves the worst-case prediction. The defect is a worst-case channel-simulation error, however, not a communication capacity or an achievable signalling rate.

For example, in a qubit collision with $\sin \phi = 4/5$, retaining the incoming visible state corresponds to $\lambda = 1$ and has worst-case error $2 \sin \phi = 1.6$. The physical root of Eqs. (17)–(18) is $\lambda_* = 0.339610\dots$, for which the error is $1.415252\dots$. Thus the minimax one-step predictor keeps only about one third of the incoming Bloch vector; this is the operational gain represented by the noise-activation transition.

The result is limited in three important ways. First, Eq. (1) is a pre-existing signalling measure, not a newly defined quantity. Second, the exact CNOT and SWAP endpoints are prior results [2]; the contribution here is the continuous-family evaluation. Third, for a generic two-qubit interaction in the canonical Cartan parametrization [15], $e^{-i(\alpha XX + \beta YY + \gamma ZZ)}$, a closed formula remains open. Joint-Pauli symmetry reduces the latter problem to optimization over the Pauli-channel tetrahedron, but the three leakage sectors interfere in trace norm and the partial-SWAP $U(2)$ symmetry is absent. In addition, the strong partial-SWAP regime is solved as an explicit scalar convex minimization rather than by a generic radicals formula for its minimizer.

7 Reproducibility

The accompanying repository separates three levels of checking, which differ in evidential status and should not be conflated. Level one is a dependency-free Gaussian-rational suite that verifies residual identities, physical witnesses, channel constraints, boundary values, and exact substitutions into the stated formulas. These are regression certificates: they do not independently prove the general block spectra, and no theorem depends on them. The focused suite is run from the repository root with

```
python3 -m unittest discover -s tests -p 'test_aqc_*.py' -v
```

The larger symbolic identities in Theorem 4 are level two. As stated in Remark 2, these are *not* optional: the sheet exclusion, resultant factorization, cofactor positivity, and Sturm count that complete the proof of Theorem 4 are performed and checked in exact rational arithmetic by the first script below, which is part of the proof of record. The word “optional” applies only to the extra Python dependency, not to the evidential role. The second script is an independent reconstruction, in the sense of Remark 1.

```
python3 -m pip install -e '[proof]'
python3 tools/derive_xy_middle.py
python3 tools/verify_block_reductions.py
```

The second script constructs the unreduced physical residual Choi operators directly and checks their characteristic polynomials against the partial-SWAP spectrum for $d = 2, 3, 4$ and the XY scalar-plus- M_3 reduction. It does not import the certificate package. The defect and optimizer phase-diagram data are generated without third-party Python dependencies:

```
python3 paper/generate_xy_plot.py
```

The code is a certificate and regression harness, with the single exception recorded in Remark 2: the analytic proofs do not depend on floating-point optimization anywhere, and they depend on computer algebra only for the exact polynomial eliminations behind Theorem 4. The physical-Choi reconstruction checks representative dimensions and parameters, while the all- d statement is the analytic block calculation in Lemma 2. The generic interior strong-branch minimizer is characterized by Eqs. (15)–(18), rather than encoded as a rational test value.

As an independent implementation check, the repository also contains

```
python3 -m pip install -e '[numerical]'
python3 tools/check_numerical_sdp.py
```

This optional CVXPY program does not import the exact certificate package. It optimizes jointly over an unrestricted CPTP effective channel and the dual diamond-norm SDP, then compares the unreduced numerical optimum with sample points in every analytic regime. These floating-point checks are not used as proof.

Data and code availability

This work generates no experimental data. All figure data files in the `paper/` directory are produced deterministically from `generate_xy_plot.py` using only the Python standard library, and are regenerated and compared byte-for-byte in continuous integration. The manuscript source, the exact-arithmetic certificate package, the symbolic proof scripts of Remark 2, and the independent numerical cross-check are openly available with pinned dependency versions at

<https://github.com/richorama/quantum-directional-signalling>.

The archived release DOI and immutable version identifier should be taken from the repository’s CITATION.cff file for the submitted version. Software is released under the MIT licence; the manuscript, figures, and data files are released under CC BY 4.0.

Use of AI tools

The work was developed with the assistance of AI tools for algebraic exploration, drafting, and code generation. Every theorem statement, proof, and certificate was checked by the author, who is solely responsible for all claims and errors.

8 Conclusion

Directional signalling can be evaluated sharply beyond isolated gates. The Ising formula $|\sin 2\theta|$ supplies a transparent baseline and continuous extension of the known CNOT point. The principal results are the two nontrivial optimizer transitions. In every finite dimension, partial SWAP reduces to an explicit scalar convex minimization with weak branch $2\sin\phi$, transition $(d^2 - 3)/(d^2 - 1)$, and known SWAP endpoint $2(1 - 1/d^2)$. The calculation of the XY line gives an exact three-regime curve, including the apparently new result that iSWAP reaches the global two-qubit signalling ceiling $3/2$, and a unique quartic root in the intermediate regime. These results provide controlled settings in which an optimal autonomous effective channel changes nonperturbatively with interaction strength. The axially symmetric XXZ surface is the next tractable step toward the full Cartan chamber [15].

A Prior-art search and attribution boundary

The novelty assessment was repeated on 30 July 2026 using the primary text and metadata of Refs. [1, 2], arXiv title/abstract searches, author searches for the four authors of those papers, and forward-citation queries. Search terms were combined across four groups:

1. directional signalling, causal influence, semicausal and approximately semicausal channels;
2. diamond distance to reduced, autonomous, or simulated subsystem dynamics;
3. partial SWAP, collision models, homogenization, Ising, XY, and iSWAP;
4. Cartan two-qubit gates, operator entanglement, tensor-product-structure distance, and quantum model reduction.

No independent paper was found that evaluates Eq. (1) continuously on the Ising, partial-SWAP, or XY/iSWAP families. Searches for close phrases such as “partial SWAP” with “diamond norm”, “iSWAP” with “diamond norm”, and “semicausal” with “diamond norm” found no overlapping calculation. Citation searches for Refs. [1, 2] likewise found no follow-up evaluating these families.

This negative search result is not a proof of novelty. Title and abstract indexes can miss results stated only in the body of a paper, and terminology such as autonomy, intertwining defect, model reduction, or approximate semicausality is not standardized. The precise attribution boundary used here is therefore conservative:

- the signalling quantity and its relation to causal influence are from Ref. [1];
- the exact CNOT value, exact arbitrary-dimensional SWAP value, depolarizing SWAP optimizer, and equal-dimension universal upper bound are from Ref. [2];
- the continuous-family formulas, thresholds, effective channels, and matching witnesses proved here are the claimed contributions.

The repository records the search scope and reproducible claim audit so that this assessment can be updated if closer prior art is identified.

B Audit details for the intermediate XY branch

This appendix records the boundary and saturation details used in Theorem 4. They are included to make the exact eliminations auditable without reconstructing the branch selection from numerical output.

B.1 Boundary exclusion and uniqueness

On the facet $w = 2u - 1$, Eq. (33) satisfies

$$P_A|_{A=1/2, \ell=(1-u)/2} = \frac{u(1-u)(1-C)}{2} > 0.$$

Since $P_\ell > 0$, the largest eigenvalue decreases as A approaches $1/2$; that endpoint cannot maximize the witness objective. At $A = 0$, the objective is

$$2\sqrt{(u - C^2)^2 + C^2 S^2}.$$

The maximizing A is unique by the argument in the main proof. Therefore, if $A = 0$ is active, Danskin’s theorem identifies the derivative of the maximized facet objective with the derivative

of this displayed boundary function. It vanishes only at $u = C^2$. The derivative into $A > 0$ there has the sign of $\Xi(C^2)$ from Eq. (34), namely

$$C(1 - C)S[2SC - (3C^3 + C^2 - C - 1)],$$

which is positive exactly after the first threshold, so $A = 0$ is not active at the only possible stationary channel value. Thus the middle saddle has $0 < A < 1/2$. Direct endpoint evaluation gives $f'(0^+) = -2$ and $f'(1^-) = 2S$, so also $0 < u < 1$.

For fixed u , concavity in A makes the maximizer set an interval. Any nontrivial interval would make the cubic $P(\ell; A, u)$ vanish identically in A , contradicting the coefficient $-\Theta$ of A^3 . For fixed C , convexity in u makes the minimizer set an interval. A flat portion on the $A = 0$ boundary is excluded by strict convexity of the displayed square root. In the interior it would force the A -discriminant polynomial $\mathcal{R}$ in the main proof to vanish identically in u , including its leading coefficient $(1 - C)^2 - 8\ell$. This contradicts $4\ell \geq 2SC > 1/2$ from Eq. (28). Hence both optimizers are unique, and Berge's theorem gives their continuity in C .

B.2 Multiplier elimination and saturation ledger

The zero-multiplier calculation starts from the unrestricted characteristic polynomial and the equations

$$P = 0, \quad P_A = 0, \quad P_u + 2P_w = 0, \quad G = 0.$$

The physical interior permits saturation by

$$A, B, u, 1 - u, C, 1 - C^2, \ell, P_\ell, \Psi.$$

Every factor is nonzero: the boundary factors were excluded above; $0 < C < 1$ and $\ell > 0$ hold in the middle interval; $P_\ell > 0$ follows from the simple positive eigenvalue; and $\Psi > 0$ follows from its negative discriminant. The substitution $B = (u - C)\ell/\Psi$ therefore loses no physical solution. The remaining generic component $u = -2\ell$ is incompatible with $u, \ell > 0$. After these exclusions, the saturated elimination ideal in $\mathbb{Q}[C]$ is generated by $5C^3 + C^2 + 3C - 1$. It vanishes only at θ_2 , establishing the nonvanishing multiplier used in the KKT argument. The optional symbolic script also prints the unsaturated resultant factors and the justification for excluding each one.

B.3 Rational enclosure of the threshold interval

The elimination bounds below use a rational window rather than decimal approximations. In the tangent variable, the first and second threshold polynomials are

$$p_1(t) = t^6 - 2t^5 - 3t^4 + 7t^2 + 2t - 1, \quad p_2(t) = t^3 + t^2 - 1.$$

Exact evaluation gives

$$p_1(1/4) = -\frac{311}{4096} < 0, \quad p_1(27/100) = \frac{31874409089}{10^{12}} > 0,$$

and

$$p_2(3/4) = -\frac{1}{64} < 0, \quad p_2(19/25) = \frac{259}{15625} > 0.$$

Using the uniqueness of the positive threshold roots therefore gives

$$\frac{1}{4} < \tan \theta_1 < \frac{27}{100} < \frac{3}{4} < \tan \theta_2 < \frac{19}{25}.$$

This both justifies the enclosing interval $[1/4, 19/25]$ and supplies the $t < 27/100$ bound used in the weak-branch principal-minor estimate.

B.4 Physical sheet, resultant, and root isolation

Put $a = 2A$ and $F = 64(1+t^2)^2 P(\Delta/4; a/2, u) = P_2 u^2 + P_1 u + P_0$. At the saddle $F_u = 0$, and direct factorization gives

$$\text{disc}_u F = 4W_+(a, \Delta, t)W_+(a, \Delta, -t),$$

where

$$W_+ = (1+t^2)^2 \left[a \left(\Delta + \frac{4(1-a)t^2}{1+t^2} \right)^2 - 2\Delta^2 + \frac{8\Delta(1-a)t}{1+t^2} \left(\frac{2at^2+1-t^2}{1+t^2} \right) \right].$$

With $\sigma = t^2/(1+t^2)$ and $b = 1-a$, the conjugate sheet obeys the exact rearrangement

$$-\frac{W_+(a, \Delta, -t)}{(1+t^2)^2} = (1+b)\Delta^2 + 4b\Delta\kappa - 16(1-b)b^2\sigma^2,$$

where

$$\kappa = SC - 2a\sigma(1-S) \geq S + C - 1 > 0.$$

Using $a(1-a) \leq 1/4$, $\Delta \geq 1/2$, and $t \in [1/4, 19/25]$ gives the rational lower bound 171381/972196. Thus the conjugate sheet cannot contain the physical saddle.

The second stationary equation, after using $F_u = 0$, is $G_8(a, \Delta, t) = 0$, defined without expansion by

$$-16G_8 = (\partial_a P_2)P_1^2 - 2P_1(\partial_a P_1)P_2 + 4(\partial_a P_0)P_2^2.$$

Exact elimination gives

$$\begin{aligned} \text{Res}_a(W_+, G_8) &= -2^{36}\Delta^{12}t^{28}(1+t^2)^{10}E_1E_2^2Q_+(\Delta, t), \\ E_1 &= \Delta^2(1+3t^2) + 4(1-t^2)(1+\Delta), \\ E_2 &= \Delta(1+t^2)^2 + 2(1+t)^2(1+2t-t^2). \end{aligned}$$

Both E_1 and E_2 are strictly positive for $0 < t < 1$ and $\Delta > 0$, so the physical saddle satisfies $Q_+(\Delta, t) = 0$.

For root isolation,

$$\begin{aligned} Q_+(1/2, t) &= \frac{1}{16}(2484t^7 + 2565t^6 + 2349t^5 + 3119t^4 + 2014t^3 + 687t^2 - 155t + 5) > 0, \\ Q_+(2, t) &= -32(9t^5 + 5t^4 + 22t^3 + 12t^2 + 7t - 1) < 0 \end{aligned}$$

on $[1/4, 19/25]$, while

$$\text{disc}_\Delta Q_+ = 2^{16}t^7(1+t^2)(t^3 - t^2 - t - 1)^2 N_9(t)^3$$

with

$$N_9(t) = 27t^9 + 378t^8 + 1026t^7 + 2106t^6 + 2772t^5 + 2646t^4 + 1658t^3 + 678t^2 + 141t + 16.$$

The leading coefficient $(1+t^2)^2(t^2+t+1)$ and the discriminant are nonzero there, preventing roots from entering, leaving, colliding, or escaping to infinity. The exact Sturm variation difference at $t = 1/2$ is one between $\Delta = 1/2$ and $\Delta = 2$.

Finally, substituting the weak and strong values into Q_+ produces, respectively, the squared threshold factors

$$p_1(t)^2, \quad p_2(t)^2,$$

with strictly positive cofactors on the enclosing window. Hence the unique physical root joins both closed branches exactly.

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